

ITU Artificial Intelligence & Data Engineering Department
YZV 202E Optimization for Data Science, Spring 2026
Term Project

Project Team Registration Deadline: April 3, 2026 11:59pm
Project Topic Proposals Deadline: April 17, 2026 11:59pm
Second Iteration Deadline: April 24, 2026 11:59pm

All students are required to read the entire project document from beginning to end.

1 General Project Description

You will apply appropriate design methods to develop a solution or prototype based on one or more selected optimization techniques for a real-life problem that relates to public health, safety, welfare, or environmental, cultural, social, and economic factors. Define a problem suitable for the dataset you have selected from the datasets listed in Section 4.

2 Group Work and Collaboration

- The project will be completed in groups of three.
- You should **write a proposal** with the **given IEEE template**. Define your problem, dataset, candidate model(s), optimization technique(s) and explain in the proposal form. The proposal form should be **maximum 2 pages** long (Excluding references). **Proposals and Reports generated by LLMs (e.g. Chatgpt, Gemini, local etc.) will not be evaluated.** Copy-paste LLM proposals are one of the top reasons for rejection in the first iteration! Failure to obtain acceptance in the second iteration will result in a **grade of VF**. If you are unsure of your project idea or proposal, you can come and speak to us beforehand to make sure you understand the requirements completely.

By the deadline, you must register your project title and team. Any student who could not form a team before the deadline, will be assigned to one of the randomly created teams.

3 Example Project Idea

Resource Allocation Problems

Use pure optimization perspective to solve a resource optimization like real-world problem. Here is an example: An inevitable earthquake is looming over Istanbul, with experts predicting its occurrence within the next six years. The government and relief organizations will need to efficiently distribute food, water, medicine, and other essential supplies to the affected districts while considering transportation challenges and fuel shortages.

There are multiple distribution centers across near to Istanbul with varying supplies. The transportation costs and time between distribution centers and affected districts depend on the distance, road conditions, and availability of petrol. Supplies may be perishable, with some items having a limited shelf life.

By using those optimization methods, you can develop a solution to efficiently distribute the supplies in the post-earthquake recovery scenario in Istanbul.

Randomness can be introduced to the problem by having variable demands and supplies, reflecting the uncertainty in the actual needs and availability of resources in a post-disaster scenario.

Determine an optimal distribution plan that allocates supplies from distribution centers to the districts that will be affected, minimizing the total transportation costs and response time while satisfying the demands of each district and considering the transportation challenges and petrol shortages.

Approach to the problem

For all projects we will like to see some common patterns.

- (i) Description of the problem and related literature
- (ii) A design model of the problem. What are the design variables, what is the objective, what are constraints.
- (iii) A problem simulation. Events are generated random or with respect to some hyper-parameters.
- (iv) Data visualizations, optimization visualizations.
- (v) Interactive controls/forms where we can choose or set hyper-parameters.
- (vi) The use of real data for costs, distances, travel times, traffic, earthquake events, or city locations is required for a fully designed experiment. (Find such datasets in links in Section 4.) However, you can add some additional data as hyper-parameters. For example, firetrucks or stations in İstanbul can be a hyper-parameter next to some real world data.

You can select appropriate optimization methods that satisfy the constraints and requirements.

4 Datasets

Please review the example dataset sites listed below. You **must** choose at least one of your datasets from below sources.

1. Ulusal Akıllı Şehir Açık Veri Platformu (National Smart City Open Data Platform) <https://ulasav.csb.gov.tr>
2. İBB Açık Veri Platformu (Istanbul Metropolitan Municipality Open Data Platform) <https://data.ibb.gov.tr/>
3. WHO <https://www.who.int/data/collections>
4. NASA <https://data.nasa.gov/>
5. EUROSTAT <https://ec.europa.eu/eurostat/web/main/data/database>
6. TÜİK (Turkish Statistical Institute) <https://data.tuik.gov.tr/>

5 Presentation

- At the end of the semester, you will present your project in a 8 minutes presentation.
- Time limit is absolute and you must not exceed it. You are suggested to practice beforehand.
- A **10 point penalty (1/18)** will be applied to the presentation grade (5/18) for groups that exceed the time limit.
- **Those who do not attend the presentation will receive a grade of 0** for the project.

6 Grading

The grading will be as follows:

1. **Project Proposal (2/18):** Well-structured problem definition, dataset explanation, candidate optimization approaches-and why they would be a good fit for your problem. See Section 2 for warnings. If your proposal is not accepted in the first iteration, 1 point (1/18) will be deducted. A link for a GitHub repository **must** be shared. The project source code must be maintained with a GitHub repository. We expect continuous development (through commits) over the course of several weeks on GitHub.

2. **Final Report (5/18):** Well-structured problem description, method presentation, experimental evaluation, and conclusion.
3. **Presentation (5/18):** Clear and concise explanation of the problem, method, experiments, and results.
4. **Implementation & Analysis (6/18):** You are expected to implement a fully functional project presented as a formatted Jupyter Notebook. Project source blocks should be clean and well-organized. They also need to be self-explanatory enhanced with Markdown cells and comments which demonstrates project phases (e.g. data processing, experiments) and the reasoning behind the written code blocks. Your project will be evaluated based on:
 - (a) Data Preparation & Processing
 - (b) Model Analysis & Solutions
 - (c) Visuals and Interactivity: Adjustable hyperparameters through sliders and visualization of results.
 - (d) Extended Analysis: In the extended analysis, we expect to see further analysis beyond the basic solution. Experiments can be performed with different methods or libraries and compared for time and efficiency.

7 Deliverables

1. **PDF Report:** Prepare a report in IEEE conference format, including sections describing the problem, formal method formulation, explanation of algorithmic workflow, real-world application, experimental evaluation, and conclusion. (Your LaTeX source files also needs to be uploaded.)
2. **Jupyter Notebook:** Provide an executable Jupyter Notebook with your implementation, experiments, and adjustable hyperparameters using sliders or text boxes.
3. **Presentation:** Prepare a PDF or PPTX presentation that outlines the problem, method, experiments, and results.

8 Policy on the Use of Existing Code

You are allowed to use codes from GitHub, research papers or text books, **but there must be a new application, problem, or model that you apply the method to.** Otherwise, your proposal will be rejected during the evaluation process. If you make use of pre-existing code or resources, it is mandatory to cite the original source appropriately.

9 AI Policy

AI tools (e.g., ChatGPT or similar systems) may be used as a supplementary aid for brainstorming, clarifying concepts, or improving language. The final submission must reflect your own understanding and work. You remain responsible for all the content you submit. We expect you to be able to explain all the content you submit.